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Revision History

REV	REVISION NOTES	Date
X1	Initial	Jun 10, 2023
A	Final Release	Jul 10, 2023
B	1.Change 3pin solder pads. 2.Update schematic per Ver.A test result.	Sep 15, 2023

FRDM-MCXN947

<Core Design>

Microcontroller Product Group
6501 William Cannon Drive West
Austin, TX 78735-8598

Designer: John Wu	Drawing Title: **FRDM-MCXN947**		
Drawn by: Kate Fan	Page Title: **TITLE PAGE**		
Approved: William Jiang	Size C	Document Number SCH-90818 PDF: SPF-90818	Rev B
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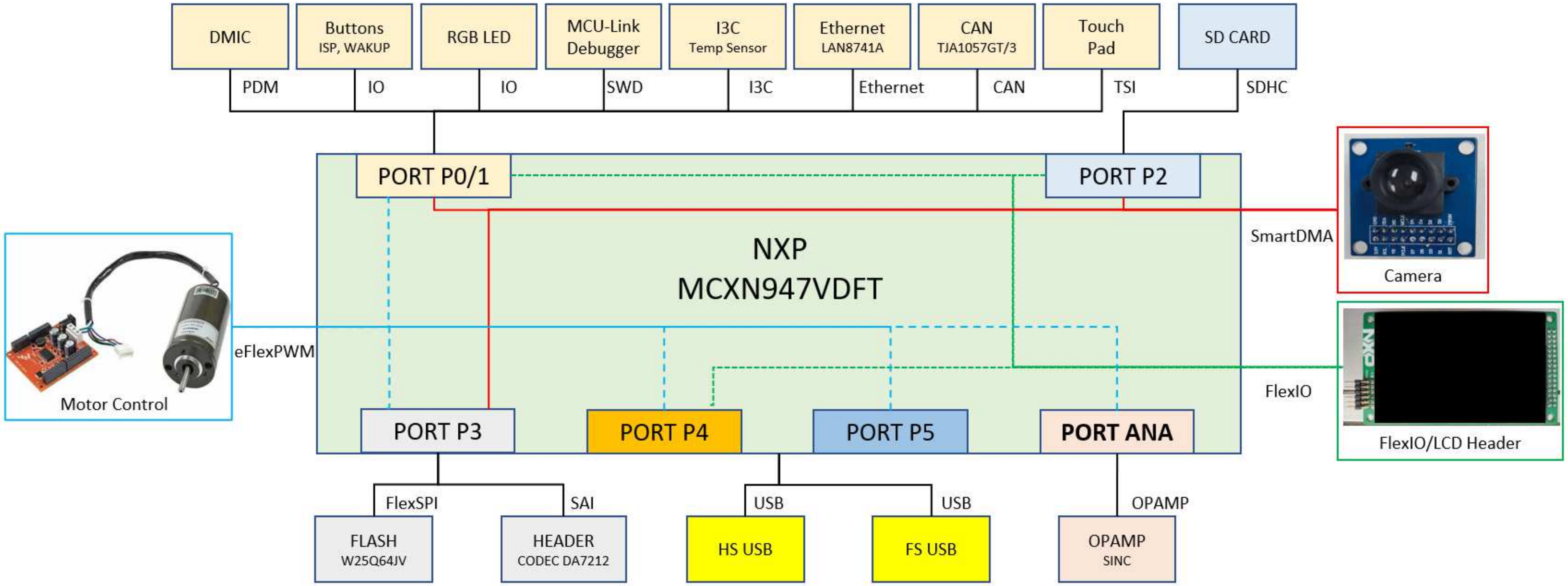
BLOCK DIAGRAM



Classification: Company Internal/Proprietary	
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SOC PORT0~2

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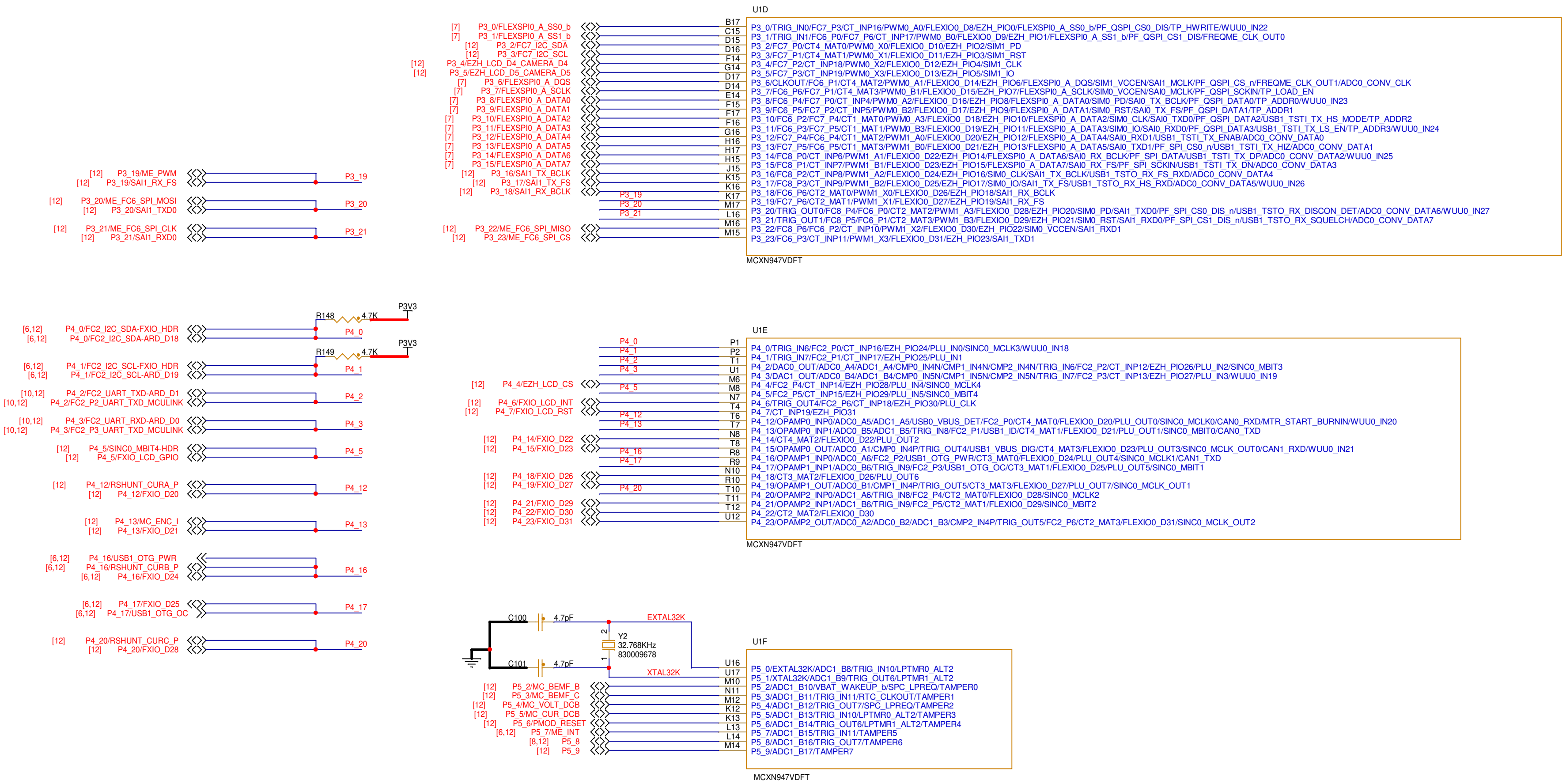
SOC PORT3~5



Classification: Company Internal/Proprietary

Drawing Title: **FRDM-MCXN947**

Page Title: **SOC_PORT3-5**

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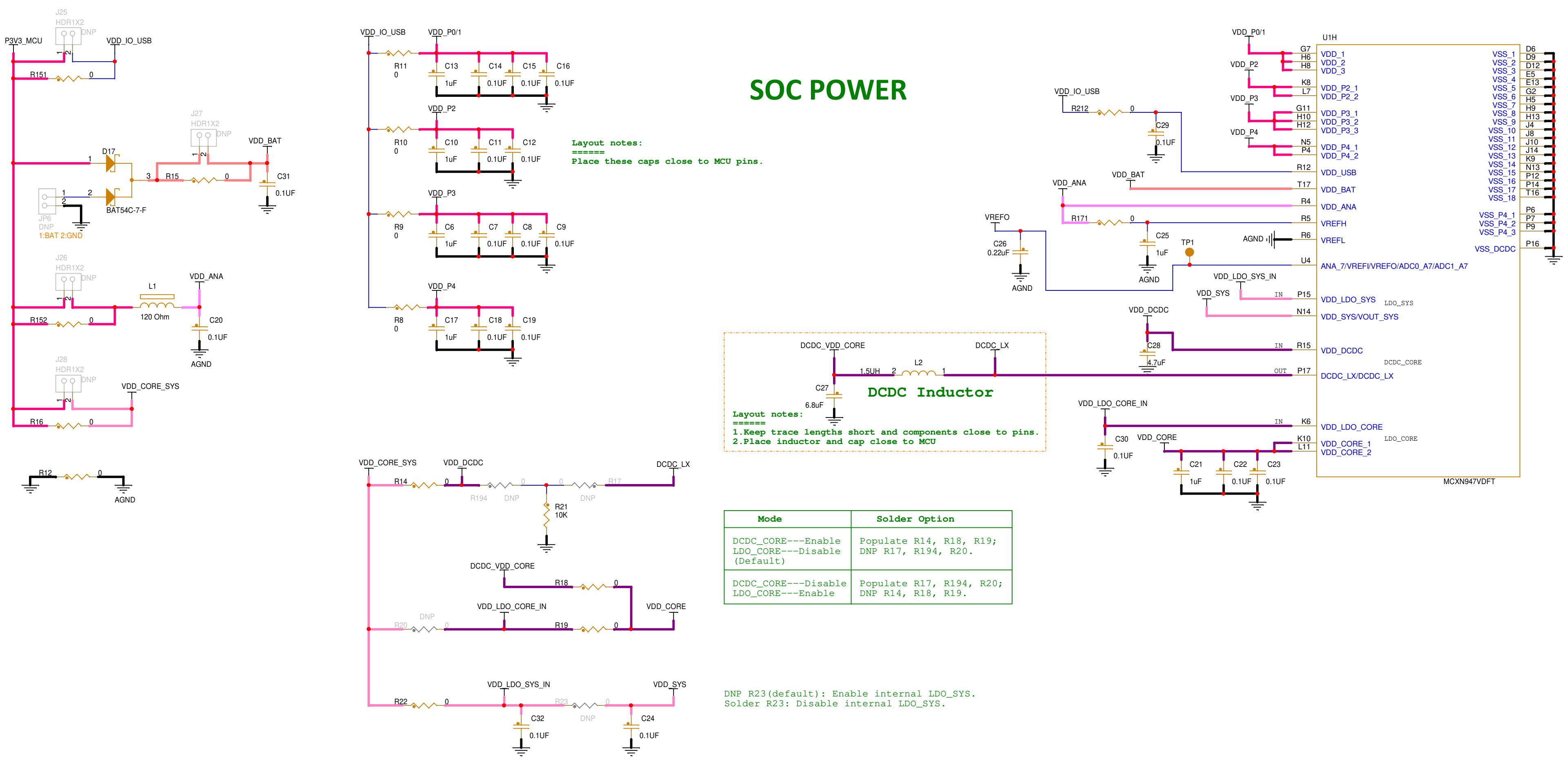
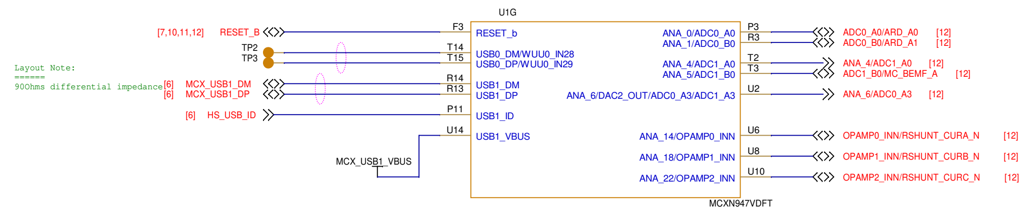
SOC POWER



Mode	Solder Option
DCDC_CORE---Enable LDO_CORE---Disable (Default)	Populate R14, R18, R19; DNP R17, R194, R20.
DCDC_CORE---Disable LDO_CORE---Enable	Populate R17, R194, R20; DNP R14, R18, R19.

DNP R23(default): Enable internal LDO_SYS.
Solder R23: Disable internal LDO_SYS.

SOC USB & Analog Signals



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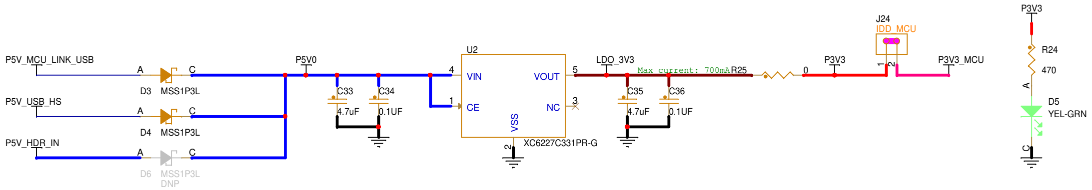
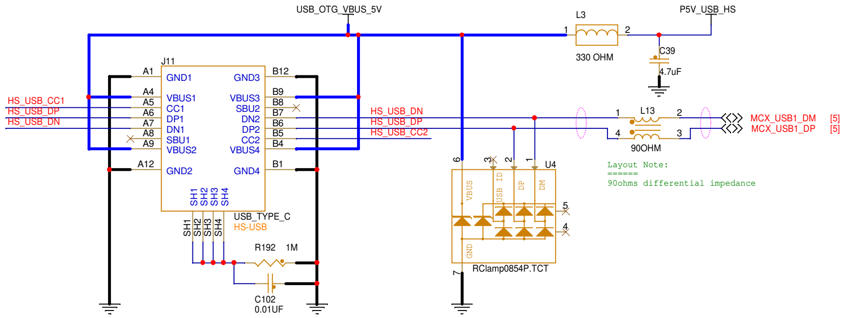
SYSTEM POWER

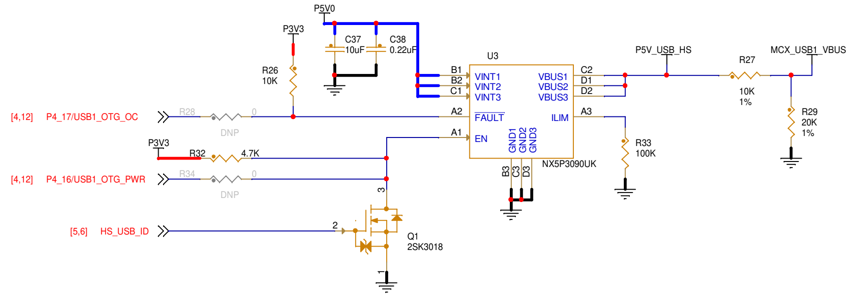


USB1_HIGH SPEED

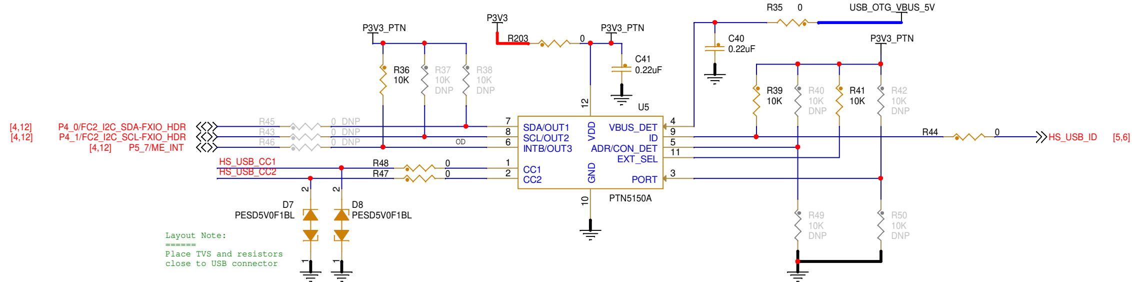
USB2.0 Type C



VBUS Power Control



CC Logic



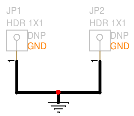
ADR/CON_DET:
When Power-up, ADR(input) function:
ADR=1: I2C Address: 0x7A(ADR)
ADR=0: I2C Address: 0x3A(ADR)
ADR=MID/FLOATING
After TINPUTLATCH, CON_DET(output) function:
CON_DET=1: Connection Detected
CON_DET=0: No Connection

EXT_SEL: External selection
High = CC1 orientation or no valid CC1/CC2 detection
Low = CC2 orientation

PORT=:
1: DFP mode
0: UFP mode
Floating: DRP mode

MOUNTING HOLE

Test Points



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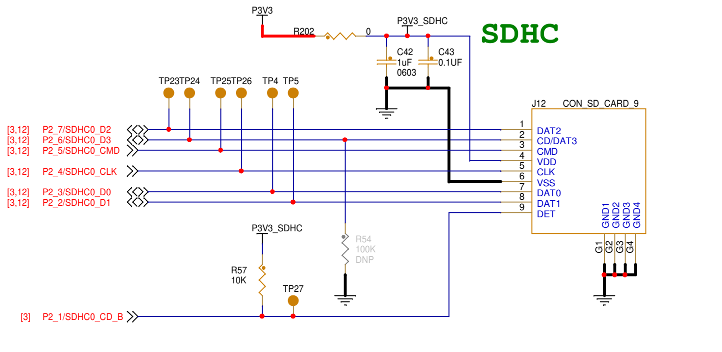
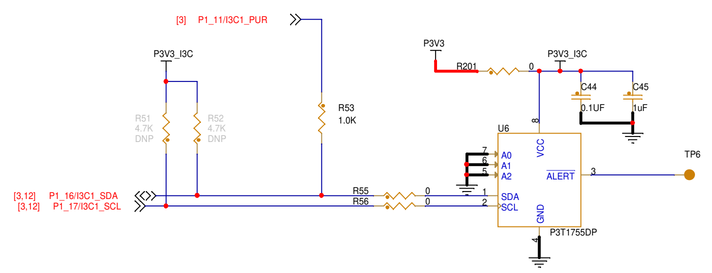
SDHC



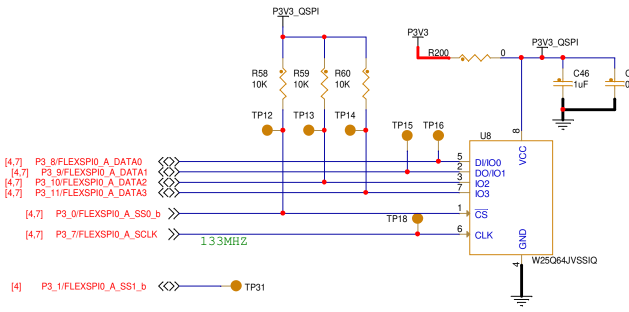
Layout Notes:
======
1) SDHC signals: equal length routing with 50ohm resistance. And as short as possible.
2) Place R54 to easily rework.

I3C SENSOR



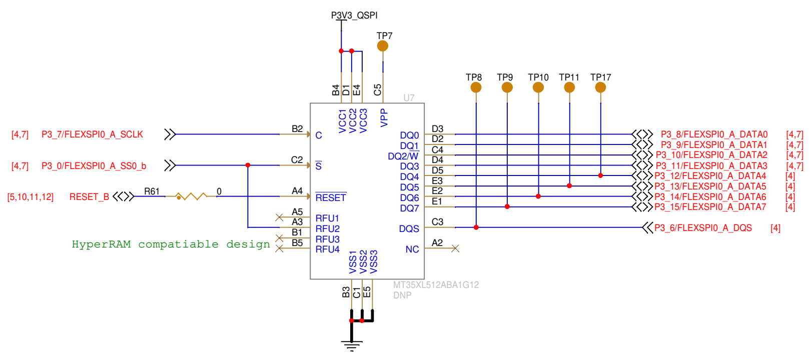
I2C ADDR: 1001000

QSPI



Other QSPI Flash Option: MT25QL128ABA1ESE-0SIT (MICRON)

Octal Flash



Layout Notes:
======
1.U7 & U8 footprint overlapped.
2.Equal length routing with 50ohm resistance.
3.Place TP31 close to TP12.

<Core Design>

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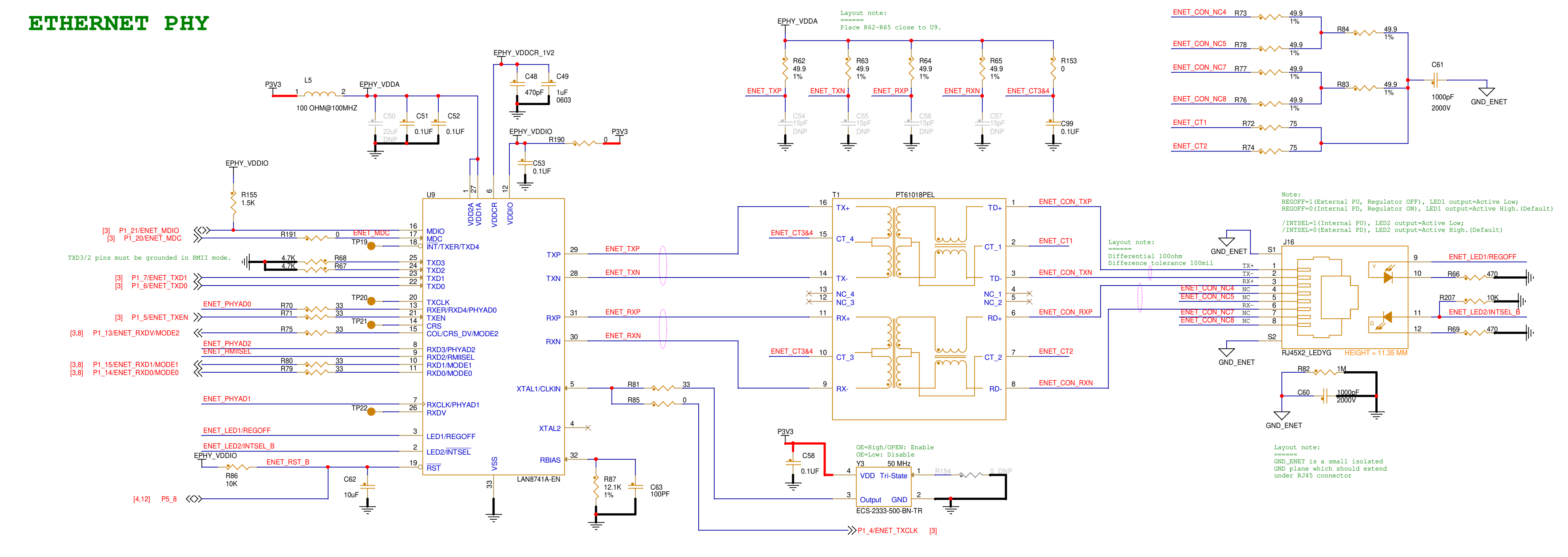
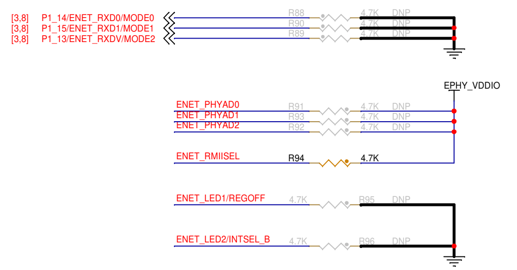
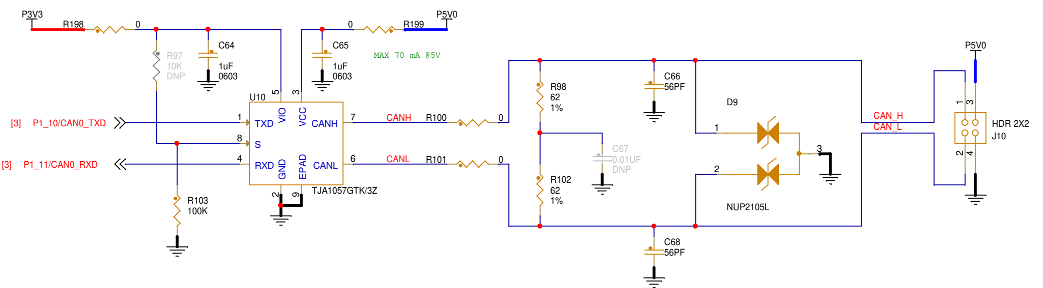
ETHERNET PHY





Config Pin (Internal State)	Description	Jumper Setting
MODE[0:2] (PU)	Mode Config 111 => All Capable, auto-neg enabled (Default)	Open (Int PU)
PHYAD[0:2] (PD)	Device Address 000 => Default	Open (Int PU)
RMIISEL (PD)	1 => RMII (Default) 0=> MII	Closed (Ext PU)
REGOFF (PD)	1 => Internal Regulator OFF 0/Floating=> Internal Regulator ON (Default)	Open (Int PU)
INTSEL_B (PU)	Floating 1 => nINT function (Default) 0=> TXD4/TXER function	Open (Int PU)

CAN PHY



Classification: Company Internal/Proprietary

Drawing Title: **FRDM-MCXN947**

Page Title: **CAN PHY & ETHERNET PHY**

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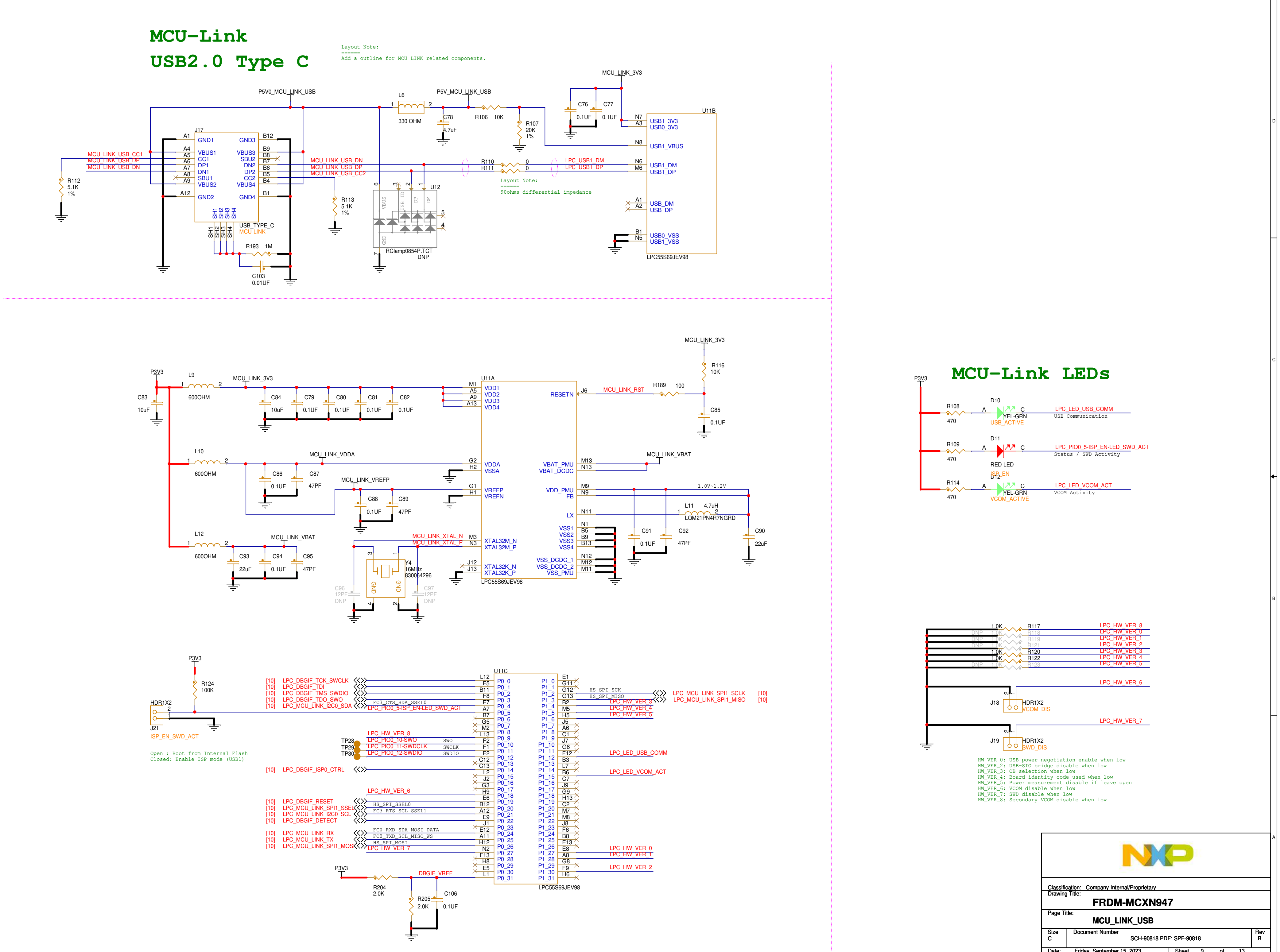

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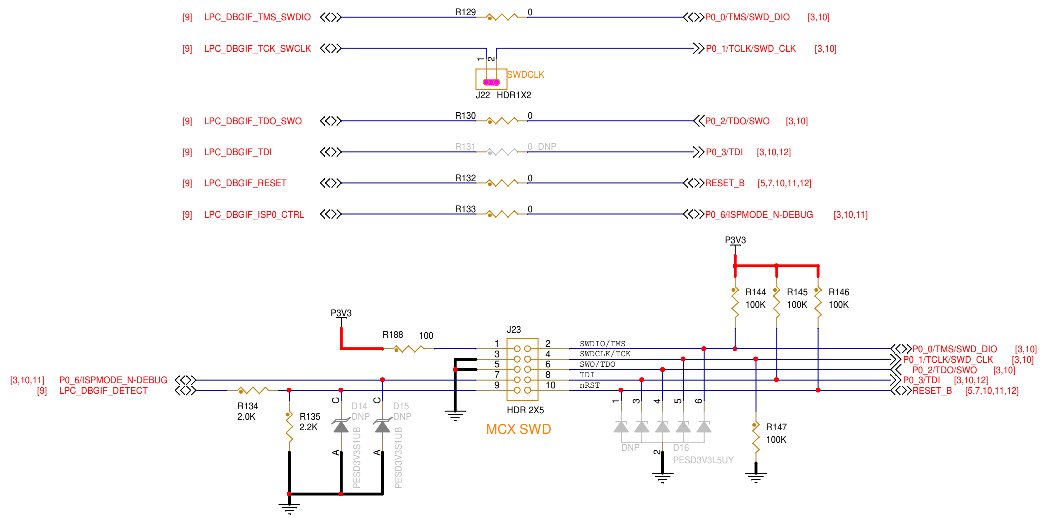
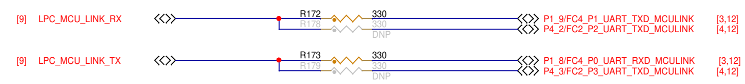
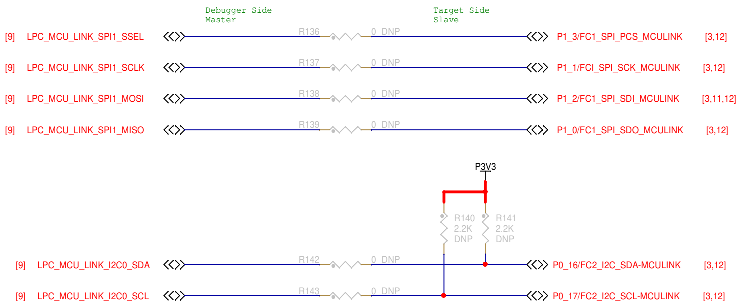
MCU-Link Debug Interface



MCU-Link UART



USB Bridge



Classification: Company Internal/Proprietary

Drawing Title: **FRDM-MCXN947**

Page Title: **MCU_LINK_DEBUG**

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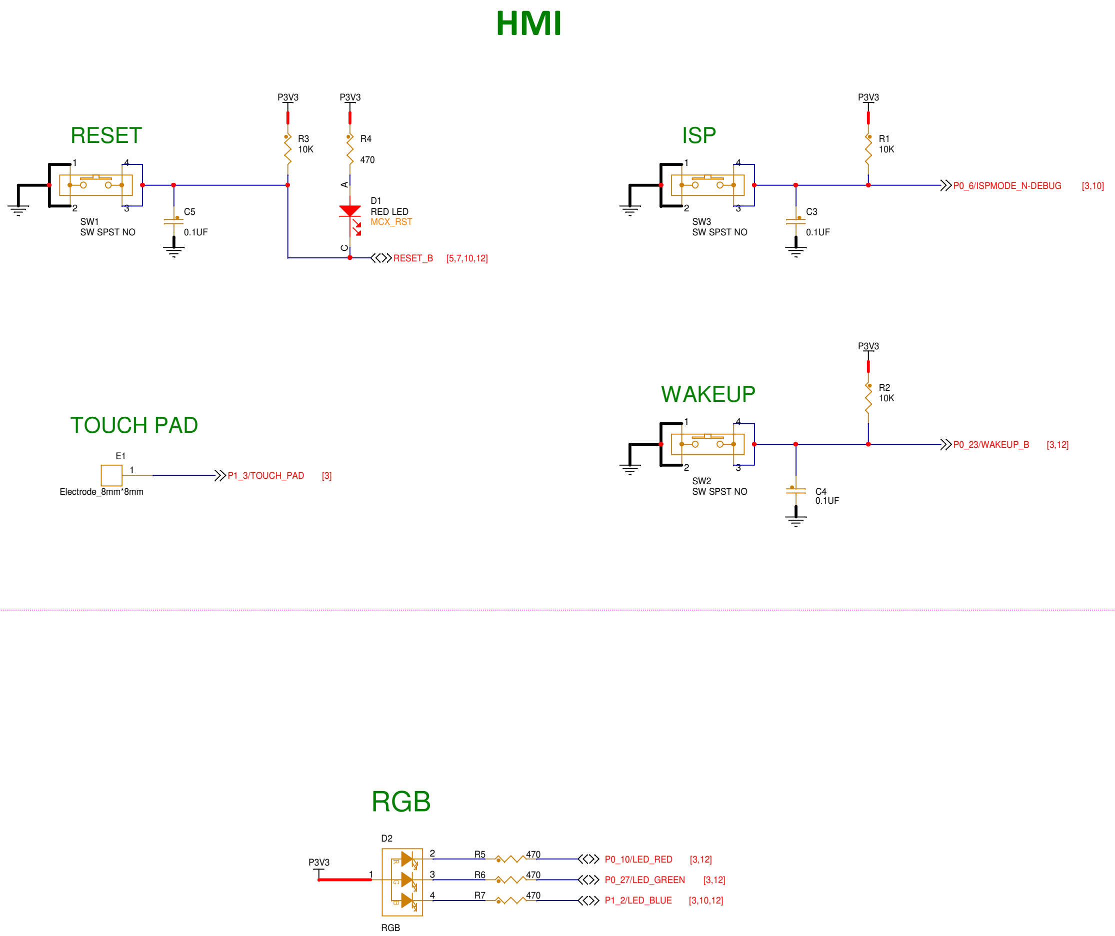

Classification: Company Internal/Proprietary

Drawing Title: **FRDM-MCXN947**

Page Title: **BLOCK DIAGRAM**

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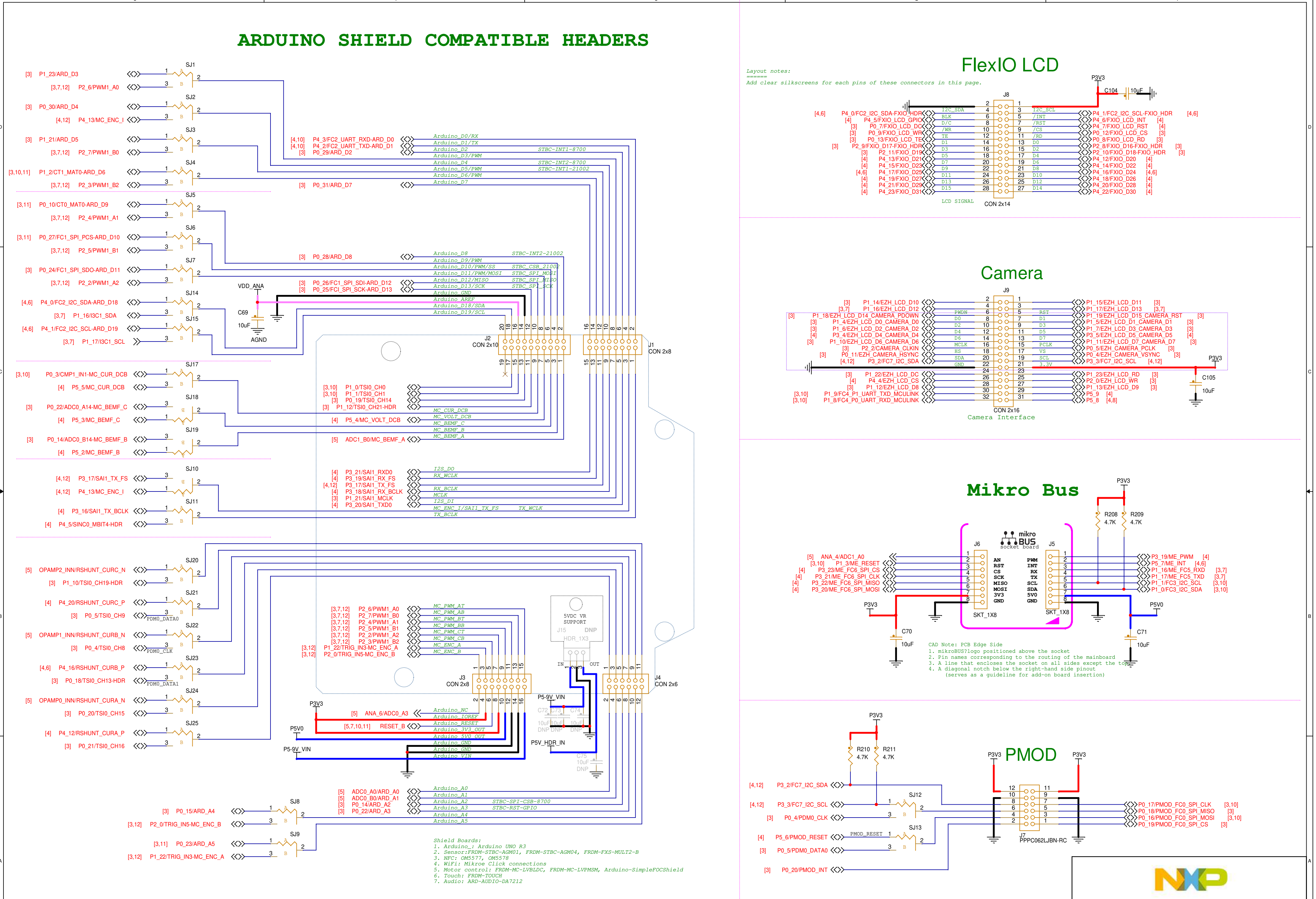

Classification: Company Internal/Proprietary

Drawing Title: FRDM-MCXN947

Page Title: HEADERS

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APPENDIX JUMPER/DNP

REF DES	JUMPER(DEFAULT)	PAGE NAME
J24	1-2	06 USB & PWR
J18,J19	OPEN	09 MCU_LINK_USB
J22	1-2	10 MCU_LINK_DEBUG

REF DES	ASSY_OPT	PAGE NAME
J25,J26,J27,J28,JP6,R17,R20, R23,R194	DNP	05 SOC_PWR
D6,JP1,JP2,R28,R34,R37,R38, R40,R42,R43,R45,R46,R49,R50	DNP	06 USB & PWR
J12,R51,R52,R54,U7	DNP	07 SD & QSPI & SENSOR
C50,C54,C55,C56,C57,C67,R88, R89,R90,R91,R92,R93,R95,R96, R97,R154	DNP	08 CAN PHY & ETHERNET PHY
C96,C97,R118,R119,R121,R123, U12	DNP	09 MCU_LINK_USB
D14,D15,D16,R131,R136,R137, R138,R139,R140,R141,R142, R143,R178,R179	DNP	10 MCU_LINK_DEBUG
C72,C73,C74,C75,J7,J15	DNP	12 HEADERS

REF DES	SHORT(DEFAULT)	PAGE NAME
SJ16,SJ26,SJ27	1-2	03 SOC_PORT0~2
SJ1,SJ2,SJ3,SJ4,SJ5,SJ6,SJ7, SJ10,SJ11,SJ12,SJ13,SJ14, SJ15,SJ17,SJ18,SJ19,SJ20, SJ21,SJ22,SJ23,SJ24,SJ25	1-2	12 HEADERS